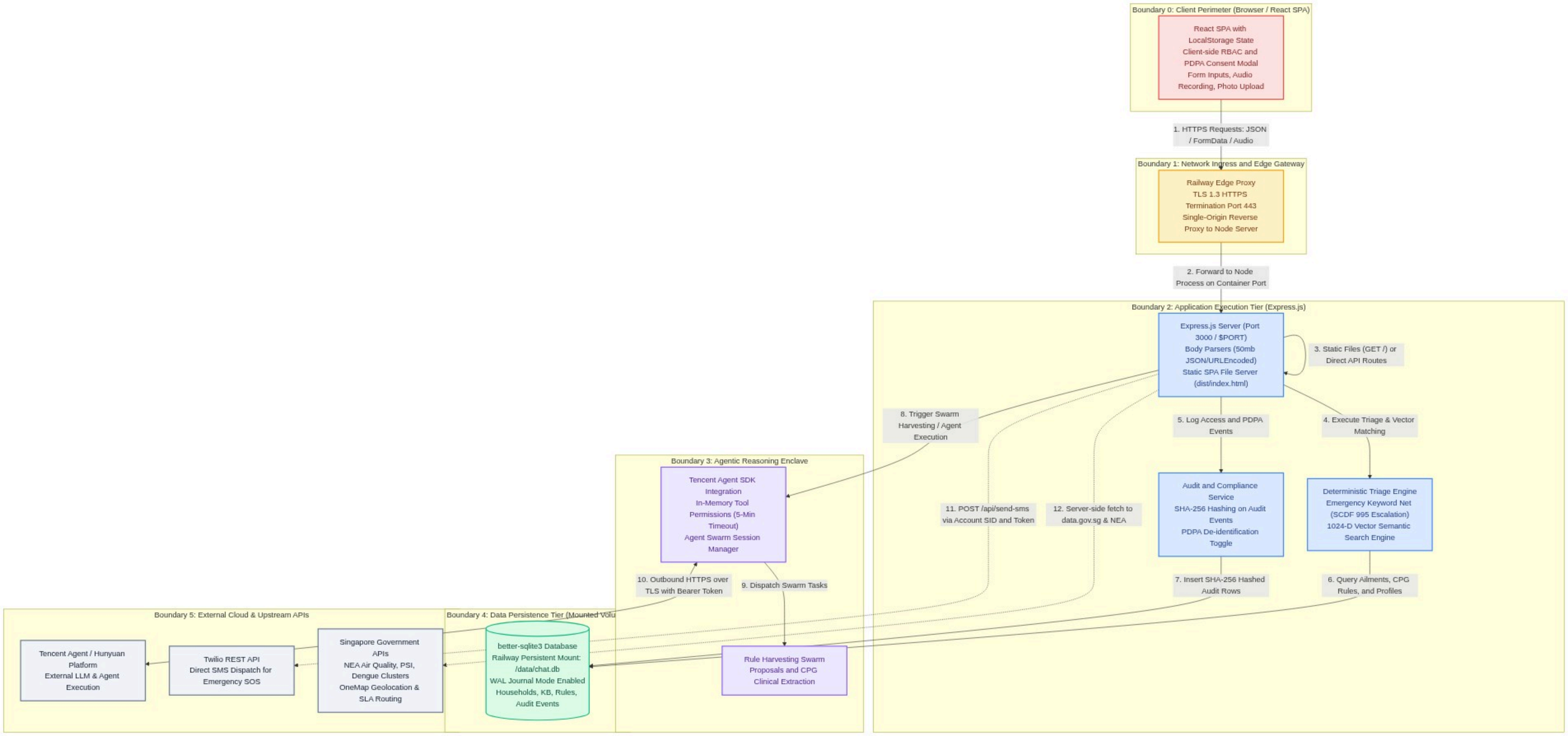

Boundary 0: Client Perimeter (Browser / React SPA)
React SPA with LocalStorage State Client-side RBAC and PDPA Consent Modal Form Inputs, Audio Recording, Photo Upload
1. HTTPS Requests: JSON / FormData / Audio
Boundary 1: Network Ingress and Edge Gateway
Railway Edge Proxy TLS 1.3 HTTPS Termination Port 443 Single-Origin Reverse Proxy to Node Server
2. Forward to Node Process on Container Port
Boundary 2: Application Execution Tier (Express.js)
Express.js Server (Port 3000 / $PORT) Body Parsers (50mb JSON/URLEncoded) Static SPA File Server (dist/index.html)
3. Static Files (GET /) or Direct API Routes
4. Execute Triage & Vector Matching
5. Log Access and PDPA Events
Audit and Compliance Service SHA-256 Hashing on Audit Events PDPA De-identification Toggle
Deterministic Triage Engine Emergency Keyword Net (SCDF 995 Escalation) 1024-D Vector Semantic Search Engine
6. Query Ailments, CPG Rules, and Profiles
7. Insert SHA-256 Hashed Audit Rows
8. Trigger Swarm Harvesting / Agent Execution
Boundary 3: Agentic Reasoning Enclave
Tencent Agent SDK Integration In-Memory Tool Permissions (5-Min Timeout) Agent Swarm Session Manager
9. Dispatch Swarm Tasks
10. Outbound HTTPS over TLS with Bearer Token
Rule Harvesting Swarm Proposals and CPG Clinical Extraction
Boundary 4: Data Persistence Tier (Mounted Volu
better-sqlite3 Database Railway Persistent Mount: /data/chat.db WAL Journal Mode Enabled Households, KB, Rules, Audit Events
11. POST /api/send-sms via Account SID and Token
12. Server-side fetch to data.gov.sg & NEA
Boundary 5: External Cloud & Upstream APIs
Singapore Government APIs NEA Air Quality, PSI, Dengue Clusters OneMap Geolocation & SLA Routing
Twilio REST API Direct SMS Dispatch for Emergency SOS
Tencent Agent / Hunyuan Platform External LLM & Agent Execution